

MODULE 1

Software Engineering-An Introduction

Software Engineering-Definition, Software life cycle activities, Challenges in System Development, Software process Models: Waterfall, Prototyping, Evolutionary, Spiral, Unified and Agile Model, Software Development Methodology.

1.1 INTRODUCTION TO SOFTWARE ENGINEERING

Software is a program or set of programs containing instructions that provide the desired functionality. Engineering is the process of designing and building something that serves a particular purpose and finds a cost-effective solution to problems.

Process:

A process is the sequence of steps executed to achieve a goal. A process is defined by cycles. Similar to a project, a process also has a beginning, middle, and end; however, this cycle repeats itself over an average period of time.

Project:

A project is defined by a fixed time, scope, and resources. When implementing a project, the goal is to execute change, usually drastic, and to incorporate that change into the day-to-day processes of the company.

Both projects and process are important for running and improving a business. However, depending on the end result trying to be achieved, one may be more suited than the other.